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Enhancing IoT Intrusion Detection Performance Using Autoencoder-Based Feature Optimization and K-Means Clustering

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Abstract

Deep learning plays a critical role in designing intrusion detection systems (IDS) to protect Internet of Things (IoT) environments against cyberattacks. However, the performance of DL-based IDS models heavily depends on the quality and balance of the training data. Real-world intrusion detection datasets often suffer from severe class imbalance, causing models to become biased toward majority attack types and underperform in detecting rare threats. While various techniques have been proposed to address class imbalance, the high dimensionality and complexity of IoT datasets remain significant challenges in building effective classifiers. Due to the dynamic nature of cyberattacks, no single method can fully address the diverse security threats in IoT networks. As a result, hybrid approaches have gained traction in enhancing cybersecurity solutions. This paper presents a novel hybrid method that combines an autoencoder and K-means clustering to generate a synthetic dataset that balances minority classes in the training set. The autoencoder reduces feature dimensionality, while K-means clustering supports oversampling of underrepresented classes. DL models are then employed for multi-class attack classification. The proposed approach is evaluated on the recent CICIoT2023 dataset. Experimental results demonstrate substantial improvements in recall, precision, and F1-score, especially in detecting minority class attacks. These findings indicate that the proposed method improves detection accuracy for rare intrusions, reduces false alarms, and supports administrators in deploying more effective IoT security measures.

Keywords: Intrusion detection; IoT; Cyberattack; Deep learning, Autoencoder; Clustering.

1. Introduction

Class imbalance is one of the most important challenges in building effective intrusion detection systems (IDS) for the Internet of Things (IoT). This issue is particularly critical in domains such as finance and healthcare, where cyberattacks can have severe consequences [1]. In these sectors, datasets typically contain an overwhelming majority of legitimate traffic and a very limited number of samples representing various attack types. This imbalance makes it

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difficult for deep learning (DL) and machine learning (ML) models to detect rare attacks, as they generally require a large volume of diverse data to learn meaningful patterns and generalize effectively.

To address this challenge, researchers have explored multiple strategies, including data-level approaches, algorithm-level modifications, and ensemble-based methods. Data-level approaches involve modifying the training dataset through oversampling, undersampling, or hybrid techniques to address class distribution before model training. In contrast, algorithm-level solutions adjust the learning process by assigning higher weights to minority classes or penalizing majority-class misclassifications [2]. Ensemble techniques combine these strategies or introduce new ensemble learning models to enhance detection performance. Despite these efforts, problems such as overfitting, information loss, increased computational cost, added noise, and difficulty handling high-dimensional data persist [3, 4, 5, 6].

In this work, we propose a novel hybrid approach that combines an autoencoder with K-means clustering to generate a synthetic dataset aimed at improving the performance of multi-class IoT intrusion detection. The autoencoder is used to reduce the high dimensionality of the dataset and remove noise, enhancing the quality of the latent space representations [7]. Specifically, the encoder compresses the original minority class data into a latent representation, while the decoder reconstructs new data with enhanced and informative features [8].

Next, K-means clustering is applied to the latent space, allowing for the identification of distinct clusters with reduced complexity compared to using raw high-dimensional data. Gaussian noise is injected into the latent vectors to maintain diversity in the reconstructed synthetic samples [9]. These processes enable more stable and effective clustering, improving the overall quality of the synthetic dataset and mitigating the class imbalance [10].

The main contributions of this paper are as follows:

- We develop a hybrid, unsupervised machine learning method that leverages autoencoders and K-means clustering to address the issue of class imbalance in IoT intrusion detection.
- Our method improves the quality and diversity of synthetic data for minority classes while reducing computational overhead and enhancing feature representation.
- We improve the generalizability of intrusion detection models by reducing bias toward majority classes and enhancing detection performance for rare attack types.
- We evaluate the performance of DL classifiers using the proposed approach on the CICIoT2023 dataset, which reflects realistic IoT network traffic.

The remaining part of this paper is structured as follows. Section 2 presents the related works. Section 3 discusses the data and methodologies in detail. Section 4 presents the experimental analysis, and Section 5 concludes the paper and highlights future work.

2. Literature Review

The growing adoption of IoT applications has been paralleled by an increase in cyber threats targeting these systems. As a result, developing effective and efficient security solutions for IoT environments has become a key focus of recent research. Numerous ML and DL-based IDSs have been proposed to enhance IoT security.

In [11], the authors proposed an anomaly detection framework that integrates K-means clustering with Contrastive Autoencoders (CAEs). The CAEs were used to compress high-dimensional network data into latent representations, while the K-means algorithm was applied to identify anomalies in the transformed space. Their method was evaluated using the NSL_KDD dataset and achieved an F1-score of 92%, outperforming standalone models in detecting network attacks.

The study in [12] focused on dimensionality reduction in financial time-series data using autoencoders, followed by various clustering techniques. The findings revealed that autoencoder-based dimensionality reduction significantly improved clustering performance compared to traditional methods, highlighting its potential in improving cluster quality for complex datasets.

In [13], the authors introduced CICIoT2023, a comprehensive real-time dataset containing real-world IoT traffic and large-scale attack scenarios. The dataset includes one benign class and 33 attack variants grouped into seven categories: DoS, DDoS, Mirai, Spoofing, Reconnaissance, Web-based, and Brute Force. A total of 105 real IoT devices were used in generating this dataset. Several ML and DL models, including Random Forest (RF), Logistic Regression, AdaBoost, Perceptron, and DL, were evaluated on binary, 8-class, and 34-class classification tasks. While RF achieved high accuracy (99.67% to 99.16%) across all classification levels, its performance on recall and precision declined in multi-class settings. The study emphasized the need for addressing class imbalance to improve detection of minority-class attacks.

Abbas et al. [14] proposed a DL-based IDS that employs Deep Feedforward Neural Network (DNN), Convolutional Neural Network (CNN), and Recurrent Neural Network (RNN) architectures to counter increasing cyber threats. Using the CICIoT2023 dataset, three variants of each model were evaluated, with RNN demonstrating the best performance: 98.61% accuracy, 98.55% precision, 98.61% recall, and 98.57% F1-score for multi-class classification. However, the study did not address class imbalance or data noise, both of which are critical concerns in real-world deployments.

The work in [15] introduced a two-step data clustering approach aimed at reducing dataset size and dimensionality to design a more computationally efficient IDS. Using the Gaining–Sharing Knowledge (GSK) optimization algorithm, they performed clustering while preserving essential information in the CICIoT2023 dataset. Classification was then performed using Multilayer Perceptron (MLP) and Autoencoder (AE) models, with MLP outperforming AE in both binary (99.26%) and multi-class (97.46%) tasks. Despite the improvement in execution time and accuracy, the study focused primarily on reducing the majority-class sample size, without addressing the scarcity of minority-class data.

In [16], a machine learning-based anomaly detection system was proposed for IoT-based healthcare environments. Models such as DNN, AdaBoost, RF, Logistic Regression, and Perceptron were evaluated on the CICIoT2023 dataset. SMOTE was used to balance the data, and RF achieved superior performance with over 99.5% across all metrics in binary classification. For multi-class scenarios (8 and 34 classes), accuracy and F1-score ranged from 95.5% to 96.32%. However, the study applied data balancing to both the training and test datasets, potentially biasing evaluation metrics and misrepresenting real-world model performance.

Resource constraints in IoT environments also make deploying IDS a challenge, necessitating lightweight solutions [17]. Efficient data preprocessing is a critical step in reducing computational costs and developing scalable IDS models [18]. The effectiveness of IDS models depends on the quality of their input data [2]. In this context, our work integrates autoencoders and K-means clustering to enhance the data preprocessing pipeline and improve the performance of multi-class attack classification in IoT environments.

Overall, prior research has addressed the challenges of imbalanced data in intrusion detection by employing various data balancing techniques and developing efficient IDS models for IoT environments. However, many of these studies have not adequately accounted for the high dimensionality and intrinsic quality of the data used to generate representative synthetic samples for minority classes. Additionally, many works overlook the risk of data leakage introduced during preprocessing, such as applying data transformations before the train-test split. These limitations can lead to overly optimistic and biased evaluations of IDS performance. To overcome these issues, we propose a novel hybrid approach that combines the dimensionality reduction capabilities of an autoencoder with the clustering strength of K-means to improve the quality and diversity of synthetic data for underrepresented classes. Importantly, data standardization was performed after the train-test split, and class balancing was applied exclusively to the training set. This methodology minimizes data leakage and ensures a more reliable and unbiased evaluation of the proposed multi-class attack classifier.

3. Data and Methodology

This section presents the methods used for dimensionality reduction, clustering, and classification in the proposed IDS model. The proposed IDS work flow presented in Figure 1.

The architecture of the proposed IDS combines an autoencoder, K-means clustering, LSTM, and DNN classifiers. The first component of the autoencoder, the encoder, compresses high-dimensional input features into a low-dimensional latent space. This compressed representation is then fed into a K-means clustering algorithm, which groups the data into seven clusters, corresponding to the minority categories defined in the dataset.

Next, the decoder component of the autoencoder reconstructs synthetic samples by taking representative cluster centroids (one per class) and introducing Gaussian noise into the latent space. This synthetic data is then used to augment the training dataset, specifically enhancing minority class representation. Finally, both DNN and LSTM classifiers are trained on the augmented dataset and evaluated using a held-out test set.

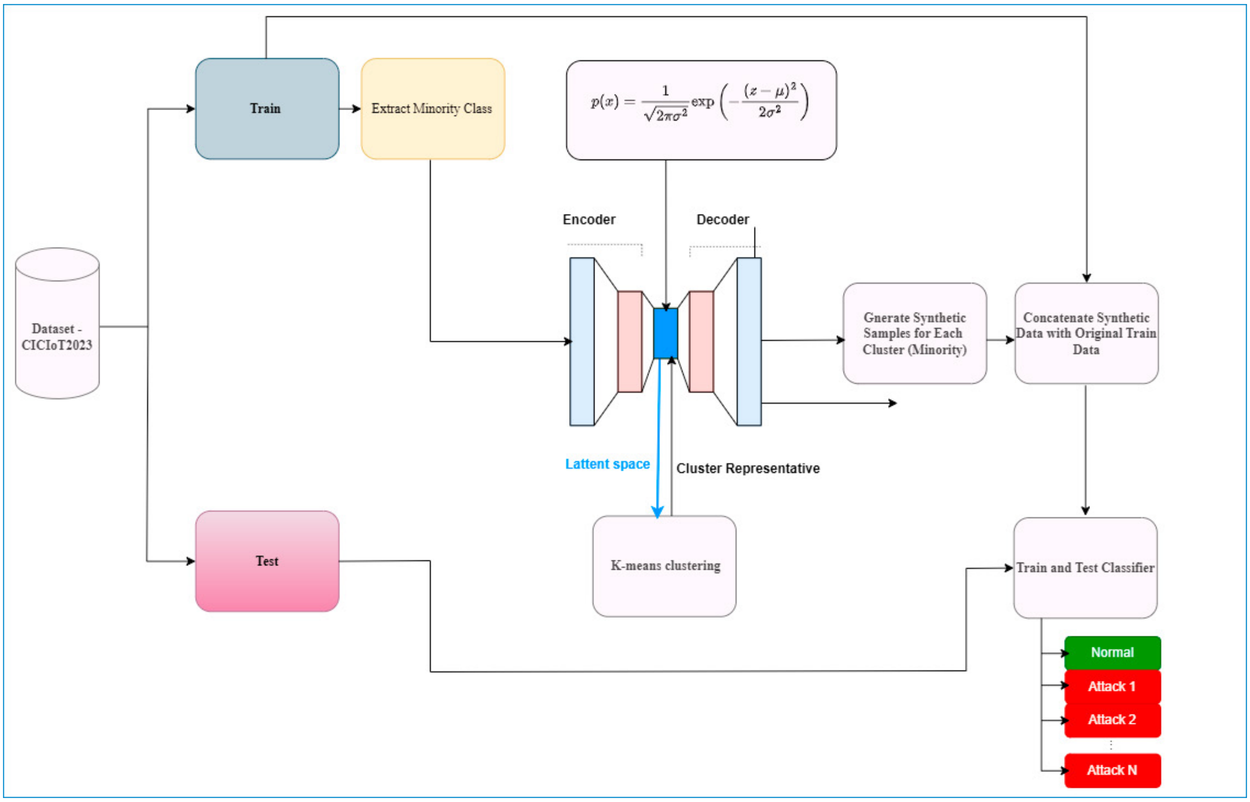


Fig. 1: Workflow of the enhanced intrusion detection model

3.1. Dataset

The effectiveness of an IDS in real-world scenarios depends heavily on the quality and representativeness of the dataset used during model development. Addressing class imbalance is a crucial step in improving the generalization and reliability of such models [19].

This study employs the CICIoT2023 dataset, a publicly available resource developed by the Canadian Institute for Cybersecurity (CIC). It includes network traffic data from 105 real IoT devices and captures one benign class and 33 distinct attack types. The dataset is available in both PCAP and CSV formats, comprising 169 CSV files designed for ML/DL evaluations.

For this work, one representative CSV file was selected, containing 232,885 instances and 47 features. In accordance with the recommendations in [13], the 33 attack types were consolidated into seven main categories: DDoS, DoS, Mirai, Reconnaissance, Spoofing, Brute Force, and Web-based attacks. The distribution of samples across these categories is shown in Table 1.

To ensure computational efficiency and enhance the model's ability to generalize to unseen traffic, the dataset was preprocessed in two stages:

Phase 1: Data cleansing and feature selection

Duplicate, constant, and highly correlated features (with Pearson correlation > 0.95) were removed. The following features were excluded during this step: *'ece_flag_number', 'cwr_flag_number', 'Telnet', 'SMTP', 'IRC', 'DHCP', 'LLC', 'Srate', 'Magnitude', 'Number', 'Radius', 'Std', 'Weight', 'ack_count', and 'rst_flag_number'*. Categorical labels were converted into numerical form using a label encoder.

Phase 2: Data splitting and transformation

The dataset was split into 80% for training and 20% for testing. After splitting, feature scaling was applied using a standard scaler to normalize the input features across both sets and prevent data leakage [20].

Following preprocessing, the minority classes in the training data were isolated and used to train the autoencoder, which then generated synthetic samples. The synthetic data generation was guided by K-means clustering and enhanced with Gaussian noise, as discussed in the following subsections. These augmented samples were added back into the training set, improving class balance.

Finally, the DNN and LSTM models were trained on the augmented dataset, and their performance was evaluated using the original test set.

3.2. Autoencoder

Feature extraction is an essential step in filtering important attributes and reducing dimensionality in high-volume datasets. The objective is to retain key information while minimizing redundancy and noise, ultimately enhancing the quality of the input for downstream models [21].

An autoencoder is an unsupervised learning algorithm designed for tasks such as dimensionality reduction and feature extraction. It consists of two neural components: the encoder and the decoder. The encoder maps the input data into a lower-dimensional latent space, while the decoder reconstructs the original data from this representation. The reconstruction aims to preserve the meaningful structure of the data with minimal loss [12].

In this work, minority-class samples are passed through the encoder to generate low-dimensional latent representations. These representations are then clustered using K-means. Representative samples from each minority class cluster are selected and injected with Gaussian noise before being passed to the decoder. This step reconstructs realistic synthetic samples, improving the diversity of the minority-class instances. The use of Gaussian noise enhances the IDS model's robustness by introducing variability that better mimics real-world attack scenarios [9]. Specifically, K-Means clustering is applied to the latent representations to identify representative centroids for each class. We then introduce zero-mean Gaussian noise drawn from $N(0, \sigma^2)$ to perturb these centroids, promoting diversity in the synthetic dataset generated by the decoder. The standard deviation σ is set via a hyperparameter ($noise\_factor = 0.1$). This ensures that the generated samples remain close to the original data distribution while introducing controlled variability to enhance diversity.

The autoencoder architecture used in this study consists of three layers. The first and last layers have 64 neurons and the latent representation layer (middle) has 20 neurons. To enhance the model's effectiveness, ReLU was used as the activation function, along with L1-L2 regularization (L1 = 0.000001, L2 = 0.00001), a dropout rate of 0.3, and the Adam optimizer with a learning rate of 0.0001. The model was trained over 60 epochs using a batch size of 64. These parameters were determined by a random search algorithm.

3.3. K-Means Clustering

K-means is an unsupervised machine learning algorithm commonly used for clustering unlabeled data based on similarity. It groups data into K clusters by minimizing intra-cluster variance. Each data point is assigned to the cluster with the nearest centroid, measured using Euclidean distance.

While effective, K-means is sensitive to the initial choice of centroids. To overcome this, we employed the K-means++ initialization method, which helps achieve better convergence and reduces the likelihood of suboptimal clustering [10]. Despite being computationally efficient [12], K-means can be negatively affected by high-dimensional data and outliers. These issues were mitigated in our work by using an autoencoder to compress the input data into a more manageable and noise-resistant latent space [12].

The parameters for K-means clustering were configured as follows: the number of clusters was set to 7 (aligned with the dataset's minority classes), using K-means++ for initialization. The algorithm was run 10 times ($n\_init = 10$) with a maximum of 300 iterations ($max\_iter = 300$), a convergence tolerance of 0.0001 ($tol = 0.0001$), and a random seed ($random\_state = 42$) to ensure reproducibility. These default parameters were obtained from the scikit-learn library.

3.4. Deep Learning Attack Classifiers

DL models are particularly effective at processing complex, high-dimensional data and identifying intricate patterns [22, 17]. In IoT ecosystems, where devices generate diverse and heterogeneous data, DL approaches are essential for designing scalable and accurate IDSs [23].

3.4.1. Deep Feedforward Neural Network (DNN)

A DNN is a class of feedforward neural networks composed of multiple hidden layers, capable of modeling nonlinear and high-level abstractions [24]. A typical DNN architecture includes an input layer, multiple hidden layers with activation functions, and an output layer. The hidden layers are responsible for extracting meaningful features and learning complex relationships from the data [17].

In this work, DNN was trained on the augmented CICIoT2023 dataset to automatically extract patterns distinguishing normal and malicious traffic. The output layer consists of eight neurons corresponding to the seven attack classes. Regularization techniques including dropout and L1/L2 penalties were applied to prevent overfitting.

3.4.2. Long Short-Term Memory (LSTM)

LSTM is a type of Recurrent Neural Network (RNN) specifically designed to address the vanishing gradient problem commonly encountered in traditional RNNs. It incorporates memory cells and gating mechanisms—input, forget, and output gates—that control the flow of information and preserve long-term dependencies in sequential data [18, 25].

Due to its ability to learn from temporal patterns, LSTM is well-suited for detecting sophisticated and time-dependent intrusion behaviors in network traffic. In our study, the LSTM model was trained on the augmented dataset to evaluate its effectiveness in capturing and classifying complex attack sequences.

Table 1: CICIoT2023 dataset: Attack class distribution

Data Classes	Records	Train	Test
DDoS	169,276	135,421	33,855
DoS	40,492	32,393	8,099
Mirai	13,322	10,658	2,664
Benign	5,400	4,320	1,080
Spoofing	2,473	1,978	495
Recon	1,742	1,394	348
Web	125	100	25
BruteForce	55	44	11

3.5. Evaluation Metrics

To evaluate the model's performance, we used four standard classification metrics: accuracy, precision, recall, and F1-score. The F1-score is a more effective metric for evaluating the performance of a multi-class intrusion detection model in the presence of unevenly distributed classes within the data [2]. The mathematical formulations and detailed explanations of each metric can be found in [17, 25]. For multi-class classification, these metrics provide insight into the model's ability to correctly identify each class instance [25]. Additionally, classification reports and confusion matrices were employed to evaluate the model's performance on a per-class basis. In this context, attack instances are treated as positive cases, while normal (benign) instances are treated as negative.

4. Experimental Analysis

4.1. Experimental Setting

The proposed model was implemented and evaluated on a laptop running Microsoft Windows 64-bit, with an Intel Core i5-12450H CPU (4.4 GHz), NVIDIA GTX 3050 4GB (GDDR6) GPU, and 16 GB of RAM. The experimental environment utilized Python 3.12.9 along with key libraries such as Keras, TensorFlow, and Scikit-Learn. Jupyter Notebook was used as the development interface.

The DNN and LSTM models were employed for attack classification using the augmented training dataset. The configuration of the DNN model includes two hidden layers and one output layer. The first and second hidden layers consist of 128 and 64 neurons, respectively, both activated using ReLU. Regularization techniques include L1 and L2 penalties (both set to 0.001) and a dropout rate of 0.2. The output layer consists of eight neurons with a *softmax* activation function. The model was compiled with the Adam optimizer (learning rate = 0.001) and trained using the *sparse_categorical_crossentropy* loss function over 70 epochs with a batch size of 32.

The LSTM model follows a similar configuration, but the LSTM layer comprises 50 memory units. It was also trained for 70 epochs with a batch size of 32 to better accommodate sequential dependencies.

4.2. Results and Analysis

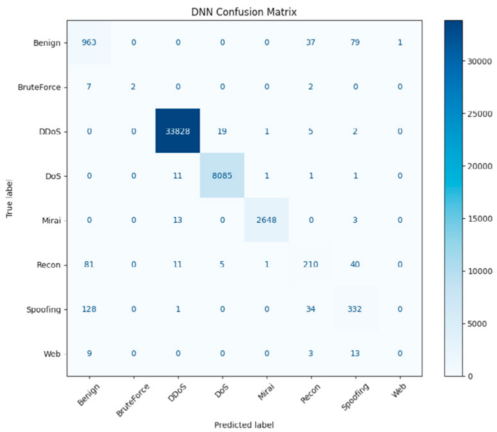
Table 2 presents the classification performance of the DNN and LSTM models on the CICIoT2023 dataset. Both models achieve high accuracy across all metrics, with LSTM slightly outperforming DNN in precision and F1-score. In addition, we conducted an ablation study to assess the effectiveness of our enhancements. First, we replaced K-Means with Random Oversampling (ROS) for both models, thereby assessing performance without K-Means-based balancing. Next, we removed the AE from the feature extraction process. The results indicate that K-Means significantly improves classification performance for DNN (confirmed by the Wilcoxon signed-rank test at $p < 0.05$), and also benefits LSTM to a lesser but still notable extent. Conversely, the performance gains from using the AE are relatively modest.

Table 2: Performance of proposed models on CICIoT2023 dataset

Model	Accuracy (%)	Recall (%)	Precision (%)	F1-score (%)
DNN+ROS	97.59	97.59	97.78	97.66
DNN+K-Means	98.86	98.86	**98.85**	98.78
DNN+AE+K-Means	**98.91**	**98.91**	**98.85**	**98.86**
LSTM+ROS	98.47	98.47	98.90	98.65
LSTM+K-Means	98.93	98.93	98.88	98.88
LSTM+AE+K-Means	**98.96**	**98.96**	**98.91**	**98.91**

Figures 2 and 3 present the classification reports and confusion matrices for the DNN+AE+K-Means and LSTM+AE+K-Means models, respectively. These visualizations provide insight into the models' ability to correctly identify each attack type. However, as shown in Figures 2(a) and 3(a), both models struggle to distinguish between benign, recon, and spoofing instances. The models classify recon and spoofing attack instances as benign, and vice versa. As well as the higher number of brute force and web attack instances misclassified as benign. These results indicate the need for further investigation into the pattern relationships between these instances and the dataset labeling.

To assess the proposed model's effectiveness, a comparison with existing models applied on the same dataset for 8-class classification is presented in Table 3. The results show that our LSTM+AE+K-Means-based model performs competitively, closely matching or surpassing prior approaches in F1-score, precision, and recall. The results also highlight that the proposed approach effectively mitigates the class imbalance problem and improves the performance of intrusion detection models compared to previous studies on multi-class classification (8 classes). Although the works presented in [13, 26] achieved higher accuracy, they were less effective at identifying minority-class attack instances, as accuracy can be biased toward the majority class in multi-class classification tasks. In contrast, our approach outperformed these models in other key metrics, demonstrating its ability to accurately classify both majority and minority attack instances in network traffic.

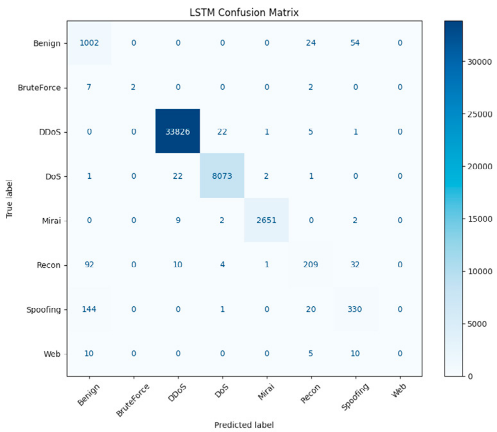


(a) Confusion matrix

	precision	recall	f1-score	support
Benign	0.8106	0.8917	0.8492	1080
BruteForce	1.0000	0.1818	0.3077	11
DDoS	0.9989	0.9992	0.9991	33855
DoS	0.9970	0.9983	0.9977	8099
Mirai	0.9989	0.9940	0.9964	2664
Recon	0.7192	0.6034	0.6562	348
Spoofing	0.7064	0.6707	0.6881	495
Web	0.0000	0.0000	0.0000	25
accuracy			0.9891	46577
macro avg	0.7789	0.6674	0.6868	46577
weighted avg	0.9885	0.9891	0.9886	46577

(b) Classification report

Fig. 2: DNN+AE+K-Means model performance on CICIoT2023 Dataset



(a) Confusion matrix

	precision	recall	f1-score	support
Benign	0.7978	0.9278	0.8579	1080
BruteForce	1.0000	0.1818	0.3077	11
DDoS	0.9988	0.9991	0.9990	33855
DoS	0.9964	0.9968	0.9966	8099
Mirai	0.9985	0.9951	0.9968	2664
Recon	0.7857	0.6006	0.6808	348
Spoofing	0.7692	0.6667	0.7143	495
Web	0.0000	0.0000	0.0000	25
accuracy			0.9896	46577
macro avg	0.7933	0.6710	0.6941	46577
weighted avg	0.9891	0.9896	0.9891	46577

(b) Classification report

Fig. 3: LSTM+AE+K-Means model performance on CICIoT2023 Dataset

Table 3: Comparison with existing works on CICIoT2023 dataset for 8-class classification

Study	Model	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
[15]	MLP	97.46	97.00	97.00	97.00
	AE	83.81	83.00	84.00	80.00
[14]	RNN	98.61	98.61	98.55	98.57
	DNN	93.77	93.77	90.03	91.75
[16]	DNN	83.07	83.07	83.63	83.00
[26]	CNN_LSTM_GRU	**99.82**	95.91	97.83	96.86
[13]	DNN	99.12	90.66	67.94	69.73
	RF	99.44	91.00	70.54	71.93
This study	DNN+AE+K-Means	98.91	98.91	98.85	98.86
	LSTM+AE+K-Means	98.96	**98.96**	**98.91**	**98.91**

4.3. Discussion

Although the proposed IDS demonstrates excellent performance, it is important to acknowledge that a high recall with relatively lower precision may lead to false positives. In practice, this results in legitimate network activities being flagged as threats, which may overwhelm system administrators with unnecessary alerts.

For example, consider a scenario where 95% of the traffic is benign and only 5% is malicious. A model achieving 99.5% accuracy might still fail to detect a significant portion of minority-class attacks, particularly if it favors the majority class. This can be misleading in sensitive domains such as healthcare and finance, where the cost of undetected attacks or false alarms can be critical [16]. Therefore, the IDS model must strike a balance between precision and recall to ensure that it detects rare but dangerous attacks while minimizing disruption from false alarms [27]. The use of data augmentation and robust feature extraction in our approach aims to support this balance and improve the practicality of the model in real-world deployments.

As shown in Table 3, the results of previous studies demonstrate that they achieved higher accuracy but lower scores on other performance metrics. In particular, the study in [26] reported an accuracy of 99.82%, outperforming our work and other studies in terms of accuracy. In contrast, our study outperforms in terms of precision, recall, and F1-score. These results highlight that the proposed approach effectively mitigates the class imbalance problem and enhances multi-class classification performance. Furthermore, the effectiveness of our model on each attack type is presented in detail in Figures 2 and 3. Note that the methods in [13] and [26] primarily apply deep networks on imbalanced or SMOTE-balanced data without addressing high-dimensionality or latent structure. Thus, our hybrid strategy is better equipped to generalize to rare attack types, as shown by the improved per-class F1-score, especially for spoofing, reconnaissance, and brute-force attacks.

5. Conclusion

This study proposed a hybrid method combining an autoencoder and K-means clustering to generate synthetic data samples, effectively balancing minority classes in the training set and enhancing IDS performance. The autoencoder reduced computational costs and suppressed noise by compressing high-dimensional input data, while K-means clustering helped identify minority-class cluster centers. These were combined with Gaussian noise and passed through the decoder to produce realistic, diverse synthetic samples.

To validate the robustness of the proposed method, our models were trained on the augmented CICIoT2023 dataset. Both models achieved superior performance compared to existing works, with DNN+AE+K-Means yielding Precision, Recall, and F1-score of 98.85%, 98.91%, and 98.86%, respectively, and LSTM+AE+K-Means achieving 98.91%, 98.96%, and 98.91%. These results demonstrate that enhancing data quality and reducing data leakage boosts IDS performance, leading to lower false alarms, faster incident response, and better risk mitigation.

The findings from this study have several practical implications for the deployment of IDSs in real-world IoT environments. First, the use of hybrid generative approaches for addressing class imbalance can substantially enhance the detection of minority-class attacks, which are often the most dangerous and least predictable. Second, the integration of dimensionality reduction and noise injection techniques promotes computational efficiency, which is crucial for deployment on resource-constrained IoT devices. Lastly, improved detection performance contributes to higher system trustworthiness and better support for network administrators tasked with managing large-scale IoT systems.

Future studies should explore the following areas to build upon the results of this work. First, it is recommended to investigate advanced techniques such as attention-based models or contrastive learning to better differentiate between overlapping or highly similar attack classes (e.g., benign vs. spoofing or reconnaissance). Furthermore, future works should assess how well the hybrid IDS model withstands adversarial attacks and incorporate defenses against data poisoning or evasion attacks. The model could also be adapted for online or incremental learning to handle streaming data in dynamic IoT environments. Finally, additional data sources such as device metadata, log files, or protocol-level features could be combined to further enrich the intrusion context and enhance detection accuracy.

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References

[1] Stephanie Ger, Yegna Subramanian Jambunath, and Diego Klabjan. Autoencoders and generative adversarial networks for imbalanced sequence classification. In *2023 IEEE International Conference on Big Data (BigData)*, pages 1101–1108. IEEE, 2023.

[2] Manahel Altalhan, Abdulmohsen Algarni, and Monia Turki-Hadj Alouane. Imbalanced data problem in machine learning: A review. *IEEE Access*, 13:13686–13699, 2025.

[3] Damien Dablain, Bartosz Krawczyk, and Nitesh V Chawla. Deepsmote: Fusing deep learning and smote for imbalanced data. *IEEE Transactions on Neural Networks and Learning Systems*, 34(9):6390–6404, 2022.

[4] John T Hancock, Huanjing Wang, Taghi M Khoshgoftaar, and Qianxin Liang. Data reduction techniques for highly imbalanced medicare big data. *Journal of Big Data*, 11(1):8, 2024.

[5] Yuxin Liu, Guangyu Du, Chenke Yin, Haichao Zhang, and Jia Wang. Clustering-based incremental learning for imbalanced data classification. *Knowledge-Based Systems*, 292:111612, 2024.

[6] Suja A Alex, J Jesu Vedha Nayahi, and Sanaa Kaddoura. Deep convolutional neural networks with genetic algorithm-based synthetic minority over-sampling technique for improved imbalanced data classification. *Applied Soft Computing*, 156:111491, 2024.

[7] Yaxin Liang, Xinshi Li, Xin Huang, Ziqi Zhang, and Yue Yao. An automated data mining framework using autoencoders for feature extraction and dimensionality reduction. *arXiv preprint arXiv:2412.02211*, 2024.

[8] Haewon Byeon, GC Prashant, Shaikh Abdul Hannan, Faisal Yousef Alghayadh, Arsalan Muhammad Soomar, Mukesh Soni, and Mohammed Wasim Bhatt. Deep neural network model for enhancing disease prediction using auto encoder based broad learning. *SLAS Technology*, 29(3):100145, 2024.

[9] Uneneibotejit J Otokwala, Andrey V Petrovskiy, and Igor V Kotenko. Enhancing intrusion detection through data perturbation augmentation strategy. In *2024 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT)*, pages 269–272. IEEE, 2024.

[10] Gutierrez-Portela Fernando, Almenares Mendoza Florina, and Calderón-Benavides Liliana. Evaluation of the performance of unsupervised learning algorithms for intrusion detection in unbalanced data environments. *IEEE Access*, 12:190134–190157, 2024.

[11] Sharmin Aktar and Abdullah Yasin Nur. Advancing network anomaly detection: An ensemble approach combining optimized contractive autoencoders and k-means clustering. In *2024 IEEE 3rd International Conference on Computing and Machine Intelligence (ICMI)*, pages 1–5. IEEE, 2024.

[12] Daniel González Cortés, Enrique Onieva, Iker Pastor López, Laura Trinchera, and Jian Wu. Autoencoder-enhanced clustering: A dimensionality reduction approach to financial time series. *IEEE Access*, 12:16999–17009, 2024.

[13] Euclides Carlos Pinto Neto, Sajjad Dadkhah, Raphael Ferreira, Alireza Zohourian, Rongxing Lu, and Ali A Ghorbani. Ciciot2023: A real-time dataset and benchmark for large-scale attacks in iot environment. *Sensors*, 23(13):5941, 2023.

[14] Sidra Abbas, Shtwai Alsubai, Stephen Ojo, Gabriel Avelino Sampedro, Ahmad Almadhor, Abdullah Al Hejaili, and Imen Bouazzi. An efficient deep recurrent neural network for detection of cyberattacks in realistic iot environment. *The Journal of Supercomputing*, 80(10):13557–13575, 2024.

[15] Hadeel Q Gheni and Wathiq L Al-Yaseen. Two-step data clustering for improved intrusion detection system using ciciot2023 dataset. *e-Prime-Advances in Electrical Engineering, Electronics and Energy*, 9:100673, 2024.

[16] Maryam Mahsal Khan and Mohammed Alkhathami. Anomaly detection in iot-based healthcare: machine learning for enhanced security. *Scientific Reports*, 14(1):5872, 2024.

[17] Qasem Abu Al-Haija and Ayat Droos. A comprehensive survey on deep learning-based intrusion detection systems in internet of things (iot). *Expert Systems*, 42(2):e13726, 2025.

[18] Shahbaz Ahmad Khanday, Hoor Fatima, and Nitin Rakesh. A novel data preprocessing model for lightweight sensory iot intrusion detection. *International Journal of Mathematical, Engineering and Management Sciences*, 9(1):188, 2024.

[19] James Halvorsen, Clemente Izurieta, Haipeng Cai, and Assefaw Gebremedhin. Applying generative machine learning to intrusion detection: A systematic mapping study and review. *ACM Computing Surveys*, 56(10):1–33, 2024.

[20] Mohamed Aly Bouke and Azizol Abdullah. An empirical study of pattern leakage impact during data preprocessing on machine learning-based intrusion detection models reliability. *Expert Systems with Applications*, 230:120715, 2023.

[21] Mohanad Sarhan, Siamak Layeghy, Nour Moustafa, Marcus Gallagher, and Marius Portmann. Feature extraction for machine learning-based intrusion detection in iot networks. *Digital Communications and Networks*, 10(1):205–216, 2024.

[22] Sidra Abbas, Imen Bouazzi, Stephen Ojo, Abdullah Al Hejaili, Gabriel Avelino Sampedro, Ahmad Almadhor, and Michal Gregus. Evaluating deep learning variants for cyber-attacks detection and multi-class classification in iot networks. *PeerJ Computer Science*, 10:e1793, 2024.

[23] Aulia Arif Wardana, Grzegorz Kołaczek, Arkadiusz Warzyński, and Parman Sukarno. Ensemble averaging deep neural network for botnet detection in heterogeneous internet of things devices. *Scientific Reports*, 14(1):3878, 2024.

[24] Vanlalruata Hnamte, Ashfaq Ahmad Najar, Hong Nhung-Nguyen, Jamal Hussain, and Manohar Naik Sugali. Ddos attack detection and mitigation using deep neural network in sdn environment. *Computers & Security*, 138:103661, 2024.

[25] Muhammad Sajid, Kaleem Razzaq Malik, Ahmad Almogren, Tauqeer Safdar Malik, Ali Haider Khan, Jawad Tanveer, and Ateeq Ur Rehman. Enhancing intrusion detection: a hybrid machine and deep learning approach. *Journal of Cloud Computing*, 13(1):123, 2024.

[26] Qawsar Gulzar and Khuram Mustafa. Enhancing network security in industrial iot environments: a deepclg hybrid learning model for cyberattack detection. *International Journal of Machine Learning and Cybernetics*, pages 1–19, 2025.

[27] Mohammed Ali Al-Garadi, Amr Mohamed, Abdulla Khalid Al-Ali, Xiaojiang Du, Ihsan Ali, and Mohsen Guizani. A survey of machine and deep learning methods for internet of things (iot) security. *IEEE Communications Surveys & Tutorials*, 22(3):1646–1685, 2020.